

Data-Driven Discovery of the Dunkle Model: A Comparative Assessment

Project Presentation

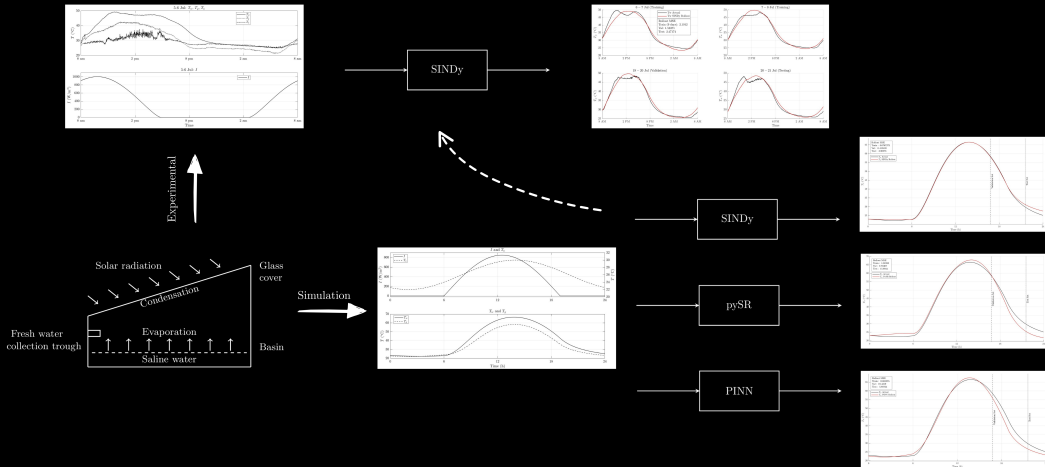
MECH 798M: Data-Driven Modeling in Science and Engineering

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Graphical Abstract

$$\frac{dT_w}{dt} = -1.266 \times 10^{-4} T_w + 1.63 \times 10^{-4} T_g - 2.937 \times 10^{-5} T_a + 1.019 \times 10^{-6} I$$



Motivation

- The Dunkle model is a canonical simulation of the thermal processes within solar stills, a desalination technology, developed in 1961 [1].
- Its derivation results in an ODE that is characteristically simple, yet mathematically convoluted.
- This project is driven by the motivation to uncover a simpler and more interpretable Dunkle model.

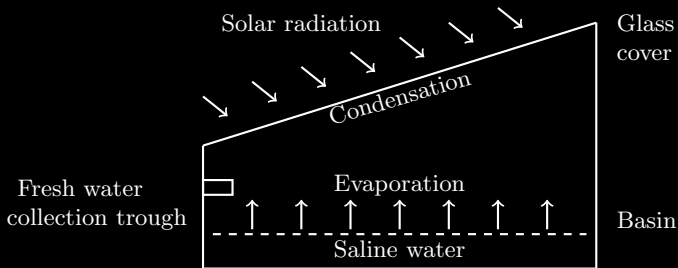


Figure 1: Single Slope Solar Still.

Review

Dunkle was able to estimate solar still yield through semi-empirical equations for convective heat flux h_c and evaporative heat flux h_e [2].

$$m = \frac{q_e}{h_{fg}} = \frac{h_e(T_w - T_g)}{h_{fg}} \quad (1)$$

where

$$h_e = 16.273 \times 10^{-3} \times h_c \left(\frac{P_w - P_g}{T_w - T_g} \right) \quad (2)$$

and

$$h_c = 0.884 \times \left[(T_w - T_g) + \frac{(P_w - P_g)(T_w)}{268.9 \times 10^3 - P_w} \right]^{\frac{1}{3}} \quad (3)$$

Review (cont'd)

Energy conservation principles applied on the various still components enable estimation of the first order quasi-linear ODE [3]:

$$\frac{dT_w}{dt} + aT_w = f(t) \quad (4)$$

Analytical solutions exist and are derived from linear assumption over infinitesimal time steps [3]:

$$T_w(t) = \frac{f}{a}(1 - e^{-at}) + T_w(t_0) \times e^{-at} \quad (5)$$

Review (cont'd)

While the ODE is characteristically simple (first order, quasi-linear), expanding its parameters unveils its convoluted nature [3]:

$$a = \frac{U_t + U_b}{m_w C_w} \text{ and } f = \frac{\alpha_{eff} \times I + (U_t + U_b) \times T_a}{m_w C_w} \quad (6)$$

$$U_t = \frac{h_t \times U_g}{h_t + U_g} \text{ and } U_b = \frac{h_w \times h_b}{h_w + h_b} \text{ where } h_w = 100 \text{ W m}^{-2} \text{ K}^{-1} \quad (7)$$

$$h_t = h_c + h_e + h_r \text{ where } h_r = \epsilon_{eff} \times \sigma \times \frac{T_w^4 - T_g^4}{T_w - T_g} \quad (8)$$

$$U_g = \frac{\frac{K_g}{L_g} \times h_g}{\frac{K_g}{L_g} + h_g} \text{ where } h_g = h_{gc} + h_{gr} = 2.8 + 3 \times v + \epsilon_g \times \sigma \times \frac{T_g^4 - T_{sky}^4}{T_g - T_a} \quad (9)$$

$$h_b = \frac{1}{\frac{L_{ins}}{K_{ins}} + \frac{1}{5.7+3.8 \times v}} \text{ and } \alpha_{eff} = \alpha'_w + \alpha'_b \frac{h_w}{h_w + h_b} + \alpha'_g \frac{h_t}{h_t + U_g} \quad (10)$$

Review (cont'd)

Plugging the expanded parameters back into the ODE:

$$\frac{dT_w}{dt} = -\frac{1}{m_w C_w} \left\{ \frac{\left[0.884 \times \left[(T_w - T_g) + \frac{(P_w - P_g) T_w}{268.9 \times 10^3 - P_w} \right]^{\frac{1}{3}} \times \left[1 + 16.273 \times 10^{-3} \times \frac{P_w - P_g}{T_w - T_g} \right] + \epsilon_{eff} \sigma \frac{T_w^4 - T_g^4}{T_w - T_g} \right] \times \frac{K_g}{L_g} \left[\frac{2.8 + 3v + \epsilon_g \sigma \frac{T_g^4 - T_{sky}^4}{T_g - T_a}}{2.8 + 3v + \epsilon_g \sigma \frac{T_g^4 - T_{sky}^4}{T_g - T_a}} \right]}{\left[0.884 \times \left[(T_w - T_g) + \frac{(P_w - P_g) T_w}{268.9 \times 10^3 - P_w} \right]^{\frac{1}{3}} \times \left[1 + 16.273 \times 10^{-3} \times \frac{P_w - P_g}{T_w - T_g} \right] + \epsilon_{eff} \sigma \frac{T_w^4 - T_g^4}{T_w - T_g} \right] + \frac{K_g}{L_g} \left[\frac{2.8 + 3v + \epsilon_g \sigma \frac{T_g^4 - T_{sky}^4}{T_g - T_a}}{2.8 + 3v + \epsilon_g \sigma \frac{T_g^4 - T_{sky}^4}{T_g - T_a}} \right]} + \frac{100 \times \frac{1}{\frac{L_{ins}}{K_{ins}} + \frac{1}{5.7 + 3.8v}}}}{100 + \frac{1}{\frac{L_{ins}}{K_{ins}} + \frac{1}{5.7 + 3.8v}}} \right\} (T_w - T_a)$$

$$- \left[\alpha'_w + \alpha'_b \frac{100}{100 + \frac{1}{\frac{L_{ins}}{K_{ins}} + \frac{1}{5.7 + 3.8v}}} + \alpha'_g \frac{0.884 \times \left[(T_w - T_g) + \frac{(P_w - P_g) T_w}{268.9 \times 10^3 - P_w} \right]^{\frac{1}{3}} \times \left[1 + 16.273 \times 10^{-3} \times \frac{P_w - P_g}{T_w - T_g} \right] + \epsilon_{eff} \sigma \frac{T_w^4 - T_g^4}{T_w - T_g}}{\left[0.884 \times \left[(T_w - T_g) + \frac{(P_w - P_g) T_w}{268.9 \times 10^3 - P_w} \right]^{\frac{1}{3}} \times \left[1 + 16.273 \times 10^{-3} \times \frac{P_w - P_g}{T_w - T_g} \right] + \epsilon_{eff} \sigma \frac{T_w^4 - T_g^4}{T_w - T_g} \right] + \frac{K_g}{L_g} \left[\frac{2.8 + 3v + \epsilon_g \sigma \frac{T_g^4 - T_{sky}^4}{T_g - T_a}}{2.8 + 3v + \epsilon_g \sigma \frac{T_g^4 - T_{sky}^4}{T_g - T_a}} \right]} \right] I \}$$

Problem Formulation

Can data-driven physics-informed machine learning techniques rediscover a simpler and more interpretable yet still generalizable Dunkle model?

Synthetic Data Description

The Dunkle model was forward modeled, with $T_w(t_0) = 23^\circ\text{C}$, $T_g(t_0) = 22.5^\circ\text{C}$ and $\Delta t = 1$ min, leading to 1,440 data points of I , v , T_w , T_g , T_a , h_c , h_e , and m .

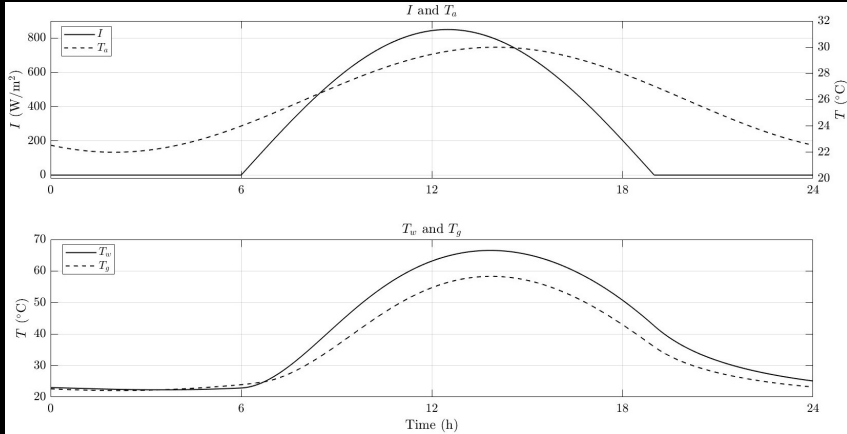


Figure 2: Synthetic Data.

Experimental Data Description

11 days of T_w , T_g , and T_a data, collected at AUB's Environmental Engineering Research Center. I data derived from Solcast API.

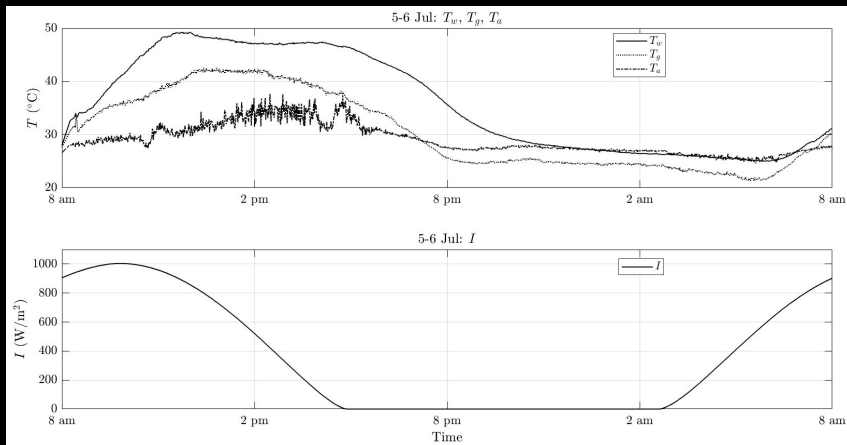
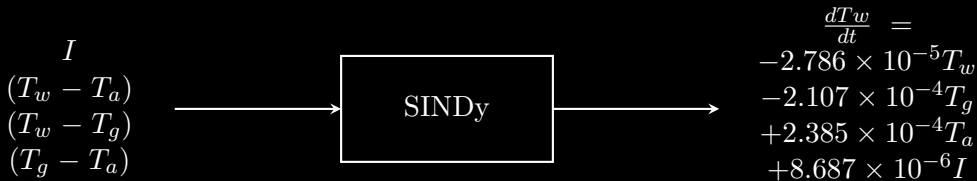


Figure 3: Experimental Data (5-6 Jul).

Methods

- Synthetic data trained using three architectures: SINDy, Symbolic Regression, and PINNs.
- The best performing architecture will then be employed for experimental data.
- Synthetic data is split sequentially into 70% training, 20% validation, and 10% testing.
- 9 days of the experimental data were utilized for training while the remaining 2 days were used for validation and testing.

Results - SINDy



$$J(\theta) = \frac{1}{N} \sum_{i=1}^N (y_i - (X\theta)_i)^2 + \lambda \sum_{j=1}^p \theta_j^2$$

$$\hat{\theta} = (X^T X + \lambda I)^{-1} X^T y$$

$$\text{Threshold sweep } (\gamma = 2.61 \times 10^{-8})$$

$$\lambda = 10^{-10}$$

Results - SINDy (cont'd)

Generalization gap = 3.89

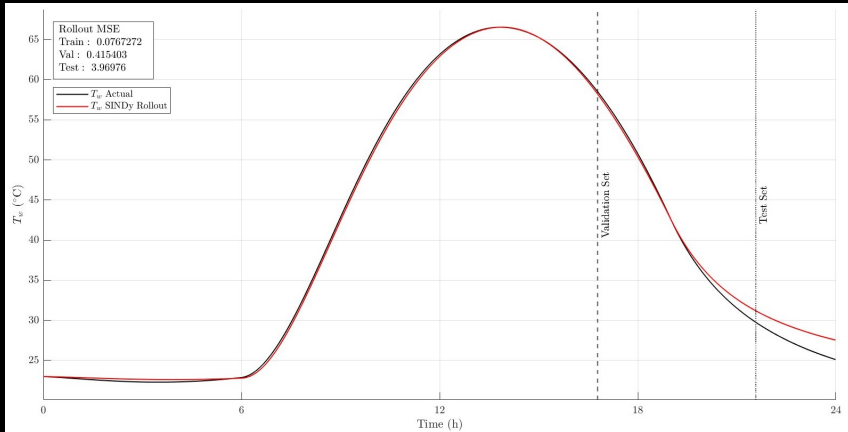
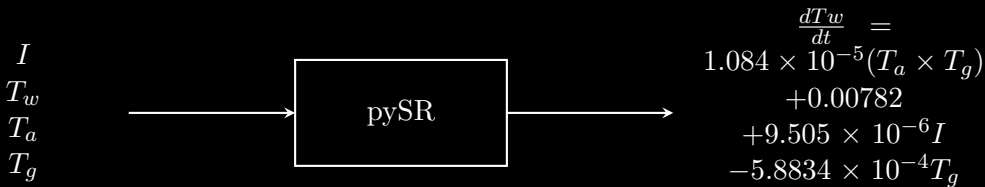


Figure 4: SINDy Predicted vs. Actual T_w .

Results - pySR



$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N \left(\frac{dT_w}{dt}_{true} - \frac{dT_w}{dt}_{pred} \right)^2$$

$$\text{Operations} = \{+, -, \times, /\}$$

200 iterations $\times$ 10 populations $\times$ 40 individuals = 80,000 candidate expressions

Results - pySR (cont'd)

Generalization gap = 14.58

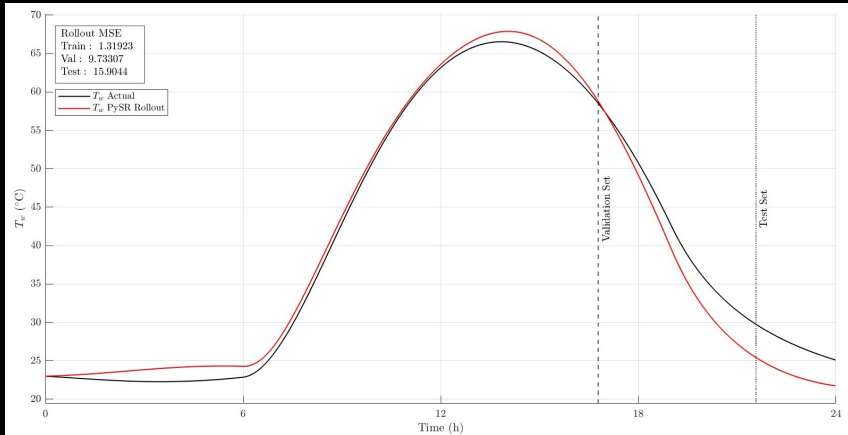
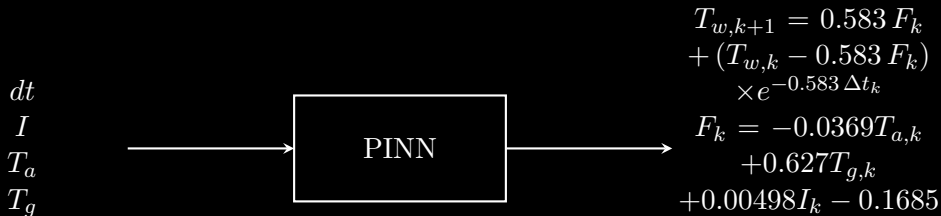


Figure 5: pySR Predicted vs. Actual T_w .

Results - PINN



$$L = w_{\text{data}} L_{\text{data}} + w_{\text{res}} L_{\text{res}} + w_{\text{ic}} L_{\text{ic}}$$

$$L_{\text{data}} = \frac{1}{N} \sum_k \left(\hat{T}_{w,k} - T_{w,k} \right)^2$$

$$L_{\text{res}} = \frac{1}{N-1} \sum_k \left(T_{w,k+1} - \left[c_1 F_k + (T_{w,k} - c_1 F_k) e^{-c_1 \Delta t_k} \right] \right)^2$$

$$F_k = c_2 T_{a,k} + c_3 T_{g,k} + c_4 I_k + c_5$$

$$L_{\text{ic}} = \left(\hat{T}_w(t_0) - 23 \right)^2$$

Results - PINN (cont'd)

Generalization gap = 5.2

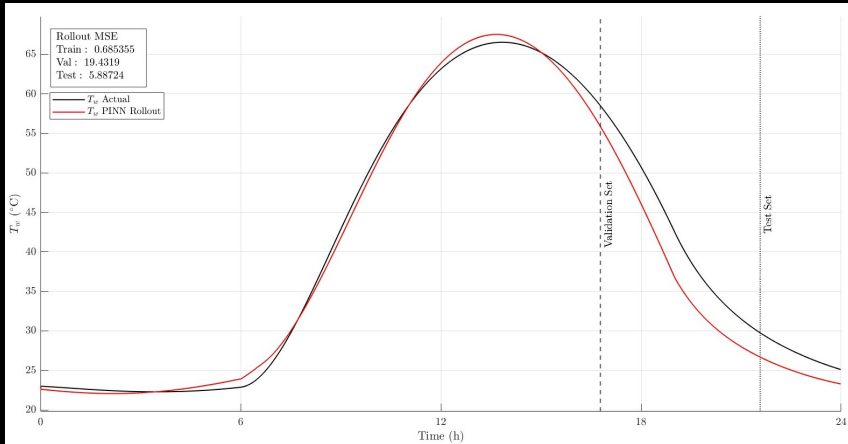
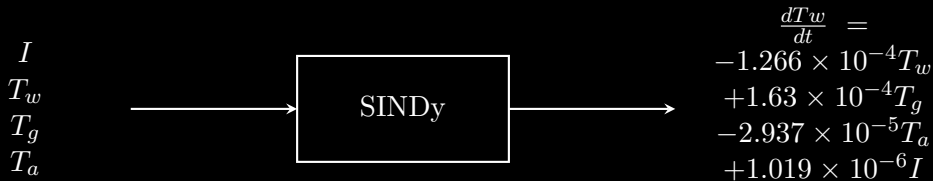


Figure 6: PINN Predicted vs. Actual T_w .

Results - SINDy Experimental



$$J(\theta) = \frac{1}{N} \sum_{i=1}^N (y_i - (X\theta)_i)^2 + \lambda \sum_{j=1}^p \theta_j^2$$

$$\hat{\theta} = (X^T X + \lambda I)^{-1} X^T y$$

$$\text{Threshold sweep } (\gamma = 1.63 \times 10^{-8})$$

$$\lambda = 10^{-10}$$

Results - SINDy Experimental (cont'd)

Generalization gap = 0.28

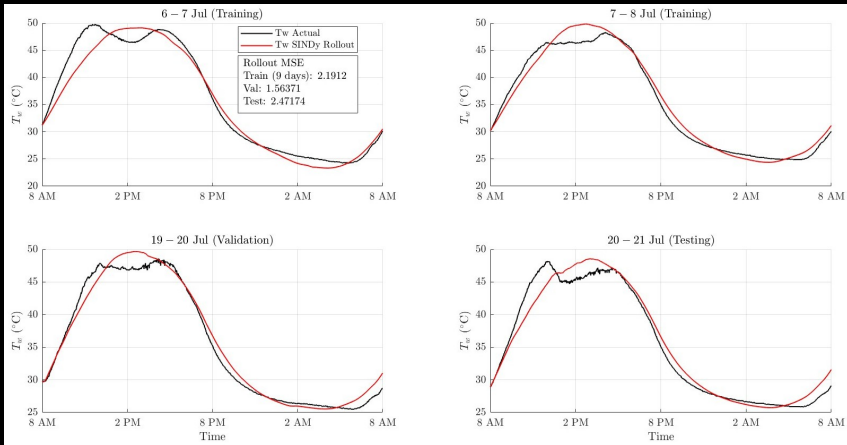


Figure 7: SINDy Predicted vs. Actual T_w (Experimental Data).

Contribution

- SINDy showcases promising potential in recovering solar still dynamics in a much simplified form, even with noisy experimental data.
- Such data-driven models can alleviate the burden of experimental research through optimizing design parameters.
- The prediction accuracy illustrates that physics-informed ML approaches could be fruitful in modeling much more complex stills.
- Future research must investigate such architectures on other types of stills, optimize design parameters within the models' constraints, and invest effort in better understanding the physics of the discovered ML models.

References

- [1] R. Dunkle, “Solar water distillation: The roof type still and a multiple effect diffusion still,” in *Proc. International Heat Transfer Conference, University of Colorado, USA, 1961*, vol. 5, 1961, p. 895.
- [2] G. M. Ayoub and L. Malaeb, “Developments in solar still desalination systems: A critical review,” *Critical reviews in environmental science and technology*, vol. 42, no. 19, pp. 2078–2112, 2012.
- [3] C. Elango, N. Gunasekaran, and K. Sampathkumar, “Thermal models of solar still—a comprehensive review,” *Renewable and Sustainable Energy Reviews*, vol. 47, pp. 856–911, 2015, ISSN: 1364-0321. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1364032115002075>.